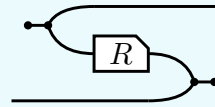


# Relation Algebra

## §3. $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

(3a)

$\circ$  is primitive, part of the data the first section lists. It turns both of  $R$ 's wires round, and the Frobenius structure **DRAWS** that — the picture and the formula below are  $R^\circ$ , not its definition:



$$R^\circ = (\circ \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circ)$$

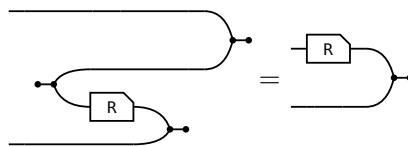
where  $\circ \triangleleft : \mathbb{I} \rightarrow a \otimes a$  opens a pair of wires out of nothing and  $\triangleright \circ : b \otimes b \rightarrow \mathbb{I}$  closes one, so the input of  $R^\circ$  is where the output of  $R$  was. Taking that formula as the definition would now be circular:  $\triangleleft$  and  $\triangleright \circ$  are  $\triangleright^\circ$  and  $\circ \triangleleft$ . The laws of  $\circ$  are that it is a contravariant 2-functor  $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ :

$$(i) \ 1^\circ = 1 \quad (ii) \ (RS)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

### §3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:

(3.1a)



$R^\circ$  is DEFINED as the bending of  $(R \otimes 1) \triangleright \circ$ , so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the  $\circ \triangleleft$  bends the  $a$  strand down and the  $\triangleright \circ$  brings it back up, while the  $b$  strand rides through untouched. Nothing else is spent below.